Confinement-loss prediction in photonic crystal fiber SPR sensors: kernel surrogates versus generative data augmentation

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Abstract

Designing photonic crystal fiber (PCF) based surface plasmon resonance (SPR) sensors requires repeated full-vectorial finite element simulation, which makes iterative geometric optimisation prohibitively slow. Machine-learned surrogates are the standard remedy, and the prevailing strategy is to grow the scarce simulation set with a generative adversarial network before fitting a deep network. We show that this strategy is counterproductive for the kernel models that suit this data regime. On a 432-sample full-vectorial finite element dataset, a Bayesian-optimised support vector regressor reaches a mean squared error of 0.121 on the held-out geometry, against 0.314 for the published GAN-augmented deep network – a 61% reduction achieved without any synthetic data. Adding the same synthetic samples to the kernel models degrades them: the support vector regressor rises to 0.197 and Gaussian process regression collapses entirely. We trace the collapse to its cause, exhibiting generated sample pairs that sit 0.011 apart in scaled feature space yet differ by 0.168 in log-scaled loss, which drives the covariance matrix towards singularity. Shapley attribution identifies the outer cladding holes as the dominant control on confinement loss, and a Monte Carlo tolerance study identifies the lattice pitch as the dominant source of resonance instability under fabrication error. The resulting surrogate is interpretable, cheap to train, and needs 70% less simulation data than comparable neural approaches.

Keywords: Photonic crystal fiber, Surface plasmon resonance, Support vector regression, Explainable artificial intelligence, Bayesian optimisation, Surrogate modelling

1 Introduction

Photonic crystal fibers coated with a thin plasmonic film form the basis of highly sensitive surface plasmon resonance sensors for chemical and biological detection. A guided core mode is deliberately allowed to leak towards a metal surface; where the mode index phase-matches the surface plasmon polariton, the confinement loss spectrum develops a sharp resonance whose wavelength tracks the refractive index of the surrounding analyte. Locating and shaping that resonance is a geometric design problem over the cladding lattice, the plasmonic film and the analyte channel.

The established way to evaluate a candidate geometry is the full-vectorial finite element method (FV-FEM), which solves Maxwell's equations on the fiber cross section. FV-FEM is accurate but expensive, and a design study sweeping several geometric parameters across a wavelength grid quickly becomes a multi-day computation. This cost, rather than any modelling difficulty, is what limits practical PCF-SPR design.

Machine-learned surrogates address the bottleneck by learning the map from geometry and wavelength to optical response, so that a trained model replaces the solver during the sweep. Chugh et al. [3] first applied artificial neural networks to PCF property computation. Deep surrogates, however, need far more training data than FV-FEM can cheaply supply, and the field's response has been to manufacture the missing data: Zelaci et al. [23] trained a Wasserstein GAN with gradient penalty on their simulation set, used it to synthesise additional samples, and reported a mean squared error (MSE) of 0.314 for the resulting augmented network.

We argue that this sequence solves a problem that need not be created. The quantity being modelled is smooth in geometry apart from a sharp resonance peak, the sample count is small ($N < 500$), and the input space is six-dimensional. These are the conditions under which margin- and kernel-based regressors are expected to beat deep networks, and recent PCF studies bear this out [11, 16]. If a kernel model is accurate enough on the original data, the augmentation step – and the generative model behind it – is unnecessary.

This paper makes that case quantitatively, and then shows the augmentation step is not merely unnecessary but harmful. Our contributions are:

1. A Bayesian-optimised support vector regression (SVR) surrogate that attains an MSE of 0.121 on an unseen geometry using only the 432 original FV-FEM samples, a 61% error reduction against the published GAN-augmented deep network and a 70% reduction in data requirement against the closest neural alternative [4].
2. An *augmentation paradox*: injecting GAN-generated samples into the same pipeline degrades SVR and destroys Gaussian process regression (GPR). We supply a mechanism rather than an observation, exhibiting generated sample pairs that are geometrically near-identical yet carry materially different targets, and showing how these drive the GPR covariance matrix towards singularity.
3. A Shapley-value decomposition of the surrogate that attributes confinement loss primarily to the outer cladding hole diameters, connecting the regression to the physics of evanescent leakage rather than leaving it a black box.
4. A Monte Carlo tolerance analysis that ranks the geometric parameters by their contribution to resonance instability under fabrication error, giving a fabrication priority that is absent from surrogate studies of this class.

A preliminary version of this work appeared in the proceedings of the 34th IEEE Signal Processing and Communications Applications Conference [8]. The present article extends it with a per-geometry cross-validation of every model, the mechanistic account and empirical evidence for the GPR collapse, the tolerance study of Sect. 5.4, and an expanded treatment of the optimisation and evaluation protocol. All figures and tables have been produced anew for this article.

Section 2 surveys related surrogate work. Section 3 describes the sensor and the FV-FEM model that generates the data. Section 4 sets out the surrogate methodology. Section 5 reports accuracy, the augmentation paradox, feature attribution, tolerance and spectral validation. Section 6 discusses limitations, and Sect. 7 concludes.

2 Related work

Neural surrogates for fiber optics.

Chugh et al. [3] established the template: sample the geometric design space with FV-FEM, fit a feed-forward network, and use it for fast sweeps. The approach is accurate where data is plentiful, and its weakness is precisely that requirement. Ehyaee et al. [4] pushed accuracy to an MSE of 0.084 on a PCF-SPR sensor, but needed 1,476 simulated samples to do it. The accuracy is real; the simulation budget is the cost.

Generative augmentation.

Zelaci et al. [23] attacked the data requirement directly, training a WGAN-GP [7] on their FV-FEM set and augmenting the training data with its output. Reported MSE improved from 0.894 to 0.314. This is the baseline we contest: the improvement is measured against a deep network that was itself data-starved, and the comparison never asks whether a model suited to small data would have needed the synthetic samples at all.

Design optimisation without deep networks.

A parallel line of work optimises PCF-SPR geometry without a deep surrogate at all. Kaziz et al. [13] combined a Taguchi design of experiments with machine learning and a genetic algorithm; Romeiro et al. [19] studied geometric optimisation of the cladding lattice directly. Goswami et al. [6] classified guided modes of a PCF-SPR sensor, and Fasseaux et al. [5] learned refractive-index dynamics for plasmonic fiber biosensors from measured rather than simulated spectra. What these share with the present work is the premise that the sample budget, not model capacity, is the binding constraint.

Kernel methods on small optical datasets.

Kalyoncu et al. [12] used k-nearest-neighbour regression for the loss characteristics of bent PCF-SPR sensors. Kaium and Mollah [11] applied GPR to PCF property estimation, and Opoku et al. [16] benchmarked ten regressors on a simulated PCF-SPR sensor, with kernel and ensemble methods leading. Applications of the same family of sensors continue to broaden [1]. These results motivate our hypothesis but stop short of testing it against a generative-augmentation baseline on identical data, which is what we do here.

Interpretability.

Khatun and Islam [14] applied Shapley additive explanations [15] to a PCF-SPR design, isolating gold film thickness as a driver of sensitivity, and Akter and Abdullah [2] argued for deliberately interpretable models in plasmonic sensor optimisation. We extend that style of analysis to a multi-family cladding lattice, where the question is not the thickness of one layer but which of several hole families controls leakage.

3 Sensor structure and numerical model

The dataset underlying this study was produced by FV-FEM analysis of the sensor cross section in Fig. 1, computed in COMSOL Multiphysics with perfectly matched layers terminating the computational domain. The structure comprises a fused-silica host, a hexagonal cladding lattice

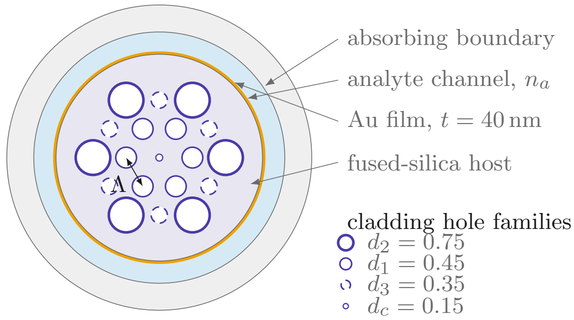

Fig. 1 Cross section of the modelled PCF-SPR sensor. Three families of air holes are arranged on a hexagonal lattice of pitch Λ in a fused-silica host, with a central defect hole of diameter d_c. Light leaking from the core couples to the surface plasmon supported by the gold film, which separates the lattice from the analyte channel. Legend diameters are in µm and give the nominal design; the dataset sweeps them over the ranges in Table 1.

of air holes, a plasmonic gold film, and an outer analyte channel that acts as the sensing medium.

Nineteen air holes in three diameter families form the cladding, together with a small central defect hole. The gold film is 40 nm thick, with permittivity taken from the Johnson and Christy measurements [9]. The refractive index of the silica host follows the Sellmeier relation [18]

$$n^2(\lambda) = 1 + \sum_{i=1}^{3} \frac{B_i \lambda^2}{\lambda^2 - C_i}, \qquad (1)$$

with λ the operating wavelength and B_i, C_i the standard Sellmeier coefficients for fused silica. The quantity to be predicted is the confinement loss, obtained from the imaginary part of the effective index as [23]

$$L_c = \frac{40\pi}{\ln(10)\,\lambda} \operatorname{Im}(n_{\text{eff}}) \times 10^4 \quad [\text{dB/cm}]. \qquad (2)$$

L_c peaks at the phase-matching condition, so it is the resonance region – where the function is least smooth – that a surrogate must reproduce.

4 Surrogate modelling methodology

4.1 Dataset and preprocessing

We use the FV-FEM dataset of Zelaci et al. [23], so that the comparison against their GAN-augmented network is made on identical data. It

Table 1 Sampled design space. Nine geometric configurations, each evaluated for three analytes over a 16-point wavelength grid, give $9 \times 3 \times 16 = 432$ FV-FEM samples.

Variable	Role	Sampled values
n_a	analyte index	1.33, 1.34, 1.35
λ	wavelength	16-point grid
Λ	lattice pitch	0.15, 0.20, 0.24
d_1	inner holes	0.25, 0.45
d_2	outer holes	0.55, 0.75
d_3	interstitial holes	0.15, 0.35
d_c	central defect	0.15 (fixed)
L_c	target	10^{-3}–59.1 dB/cm

Geometric values in µm; λ is retained in the native units of the source dataset.

contains 432 labelled samples over six input variables: analyte refractive index n_a, wavelength λ, lattice pitch Λ, and the three cladding hole diameters d_1, d_2, d_3. Table 1 summarises the sampled design space.

The target spans nearly five orders of magnitude and is dominated by the resonance peak, so we regress the log-scaled response $y = \log_{10}(L_c \times 10^8)$. This both compresses the dynamic range and makes squared error a sensible loss across the whole spectrum rather than one concentrated at the peak. Inputs are min-max scaled to $[0, 1]$, which matters for the radial basis function (RBF) kernel because an unscaled wavelength axis would otherwise dominate the distance metric.

4.2 Evaluation protocol

Random train-test splitting would be misleading here. The 432 samples are not independent: they come from nine geometric configurations, each contributing a full wavelength sweep. A random split leaves samples from the same geometry on both sides, so the model is asked to interpolate within a spectrum it has already seen, and the reported error understates what matters – generalisation to a geometry that has not been simulated.

We therefore evaluate at the level of geometries, under two protocols:

- **Fixed partition.** The 7–1–1 split of [23]: configurations 1–7 for training, 8 for validation, 9 held out for test. This reproduces the published protocol exactly and is the basis for the headline comparison.
- **Leave-one-geometry-out (LOGO).** Each of the nine configurations serves in turn as the test set, the other eight as training data. This gives nine independent estimates and, unlike the fixed partition, reports the variance across geometries as well as the mean.

The fixed partition alone is a single draw and can flatter a model that happens to suit configuration 9. LOGO is the more informative of the two, and we report both.

4.3 Support vector regression

SVR fits a function that deviates from the observed targets by at most ϵ while remaining as flat as possible, penalising only the samples that fall outside that tube [21]:

$$\min_{w,b,\xi,\xi^*} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*), \tag{3}$$

subject to

$$\begin{cases} y_i - \langle w, x_i\rangle - b \le \epsilon + \xi_i, \\ \langle w, x_i\rangle + b - y_i \le \epsilon + \xi_i^*, \\ \xi_i, \xi_i^* \ge 0, \end{cases} \tag{4}$$

where C is the box constraint and ξ_i, ξ_i^* are slack variables. We use the RBF kernel $K(x_i, x_j) = \exp(-\gamma\|x_i - x_j\|^2)$.

The relevant property for this problem is that the fit is determined by the support vectors – the samples on or outside the tube boundary – rather than by all 432 points equally. Near the resonance, where the response turns sharply, that concentrates model capacity exactly where the function is hardest, and it is why the method tolerates a small sample count. The dual formulation and the Karush-Kuhn-Tucker conditions are given in the Online Supplement, Sect. S1.

4.4 Gaussian process regression

GPR places a Gaussian process prior $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$ over functions, so that the posterior mean and covariance at a test input x_* are [17]

$$\bar{f}_* = K_*^\top \left[K + \sigma_n^2 I\right]^{-1} y, \tag{5}$$

$$\mathrm{cov}(f_*) = K_{**} - K_*^\top \left[K + \sigma_n^2 I\right]^{-1} K_*, \qquad (6)$$

with K the training covariance, K_* the train-test covariance and σ_n^2 the noise variance. We use an RBF kernel with an additive white noise term, which regularises the inversion in (5) and (6). GPR offers calibrated uncertainty, which SVR does not, and that makes its behaviour under augmentation in Sect. 5.2 diagnostic: the same smoothness assumption that yields the uncertainty estimate is what the synthetic data violates.

4.5 Hyperparameter optimisation

C, γ and ϵ interact, and grid search over three coupled continuous parameters wastes most of its budget in uninformative regions. We instead use Gaussian-process Bayesian optimisation [22] with expected improvement, 32 evaluations, and log-uniform priors over $C \in [1, 2000]$, $\gamma \in [10^{-4}, 1]$ and $\epsilon \in [10^{-4}, 10^{-1}]$, scoring each candidate by three-fold cross-validated MSE on the training partition only. The selected configuration and the full search space are tabulated in the Online Supplement, Sect. S2.

5 Results

5.1 Predictive accuracy

Table 2 reports every model under both protocols. On the fixed partition the optimised SVR reaches an MSE of 0.121, against 0.314 for the GAN-augmented deep network of [23] and 0.894 for the unaugmented network – a 61% error reduction relative to the augmented baseline, obtained without generating a single synthetic sample. GPR reaches 0.185, also well ahead of both neural baselines.

The LOGO protocol matters more, and it separates the models further. Averaged over the nine held-out geometries the SVR attains an MSE of 0.246 with a standard deviation of 0.226, against 0.825 ± 0.609 for the augmented benchmark. The gap in the standard deviation is the more useful number for a design tool: a surrogate that is accurate on average but occasionally wrong by a wide margin on an unfamiliar geometry cannot be trusted to drive an optimisation loop. Per-fold results are given in the Online Supplement, Sect. S3.

The comparison with Ehyaee et al. [4] is worth stating carefully. Their ML-enhanced ANN reaches an MSE of 0.084, better than our 0.121. It does so with 1,476 training samples against our 432. Since the training samples are FV-FEM solutions, that difference is the dominant cost in the entire pipeline, and a 70% reduction in it for a 0.037 increase in MSE is the trade a design study would take.

The mechanism behind the SVR result is the one anticipated in Sect. 4.3. A deep network fits a global function and needs enough samples to constrain it everywhere; with nine geometries that constraint is weak. SVR instead concentrates on the support vectors that bound the transition between the high- and low-loss regions, which is where the response actually carries information, and extrapolates to an unseen geometry from that boundary rather than from a globally interpolated surface.

5.2 The augmentation paradox

We next applied the augmentation strategy of [23] to our own models, generating synthetic samples with the same WGAN-GP configuration and adding them to the training partition. If synthetic data helps data-hungry models, the question is whether it helps models that were not starved to begin with.

It does not. Table 3 shows SVR degrading from 0.121 to 0.197 and GPR failing outright. The direction is consistent across the LOGO folds (Online Supplement, Sect. S3): augmentation is not adding noise that averages out, it is displacing the fit.

The GPR failure is the informative case, because it is not a gradual degradation but a numerical breakdown, and its cause is identifiable. A GAN is a probabilistic sampler with no constraint tying it to Maxwell's equations, so nothing prevents it from emitting two samples that are geometrically almost identical while carrying materially different losses. Such a pair is physically impossible – the same geometry at the same wavelength has one confinement loss – but it is entirely consistent with a generator that has learned the marginal distributions without the physics.

To confirm this we generated 50,000 samples, found each one's nearest neighbour in the scaled

Table 2 Accuracy of each surrogate under both evaluation protocols. Rows are metrics and columns are models, so that the fixed-partition result and the distribution over the nine leave-one-geometry-out folds can be read against each other for a given model. Lower is better throughout; dashes mark figures not reported in the source work.

	Published neural baselines		This work		
Metric	Deep net [23]	GAN-augmented [23]	**SVR**	GPR	ML-enhanced ANN [4]
Fixed-partition MSE	0.894	0.314	**0.121**	0.185	0.084
LOGO mean MSE	—	0.825	**0.246**	—	—
LOGO standard deviation	—	0.609	**0.226**	—	—
Training samples	432	432 + synth.	**432**	432	1,476
Synthetic data	no	yes	**no**	no	no

Table 3 Effect of adding GAN-generated samples to the training partition, split out here from the accuracy comparison because it measures a different thing: not which model is best, but what augmentation does to a model that did not need it.

Model	Real only	Augmented	Change
SVR	0.121	0.197	+63%
GPR	0.185	23.844	collapse
Deep network [23]	0.894	0.314	−65%

Fixed-partition MSE. The neural row is reproduced from [23] for contrast: the same synthetic samples that help a data-starved network harm both kernel models.

Table 4 The three most contradictory nearest-neighbour pairs among 50,000 GAN-generated samples, ranked by target discrepancy per unit of feature distance. Quantities run down the rows and pairs across the columns; each column is one pair of generated samples that are near-identical in the input space but carry incompatible targets.

Quantity	Pair A	Pair B	Pair C
Feature distance	0.0111	0.0089	0.0088
Target 1 ($\log_{10}$)	4.4466	4.3722	4.3280
Target 2 ($\log_{10}$)	4.6144	4.4958	4.4347
Target discrepancy	0.1678	0.1236	0.1067
Discrepancy/distance	15.09	13.81	12.06

Feature distance is Euclidean in the min-max scaled seven-dimensional space the generator was trained on; targets are the $\log_{10}$-scaled confinement loss.

feature space, and ranked the resulting pairs by the ratio of target discrepancy to feature distance. Table 4 lists the three worst. The leading pair sits 0.011 apart in a space scaled to the unit cube, yet its targets differ by 0.168 in log-scaled loss, a 45% discrepancy in physical units. The search protocol is detailed in the Online Supplement, Sect. S4.

For GPR this is fatal rather than merely unhelpful. The posterior in (5) requires inverting $[K + \sigma_n^2 I]$. Two inputs at distance 0.011 produce near-identical rows in K, so the matrix approaches singularity; the white-noise term σ_n^2 is what normally absorbs this, but it is calibrated to the noise level of the real data and is far too small to absorb a 0.168 jump. The inverse is then dominated by the ill-conditioned directions and the predictive variance explodes. SVR survives because its ϵ-insensitive loss simply admits both contradictory points as bounded support vectors – it fits a worse function, but it does not break.

The general lesson is that generative augmentation trades a data-quantity problem for a data-consistency one. A model that needs the quantity may accept the trade. A model that does not is strictly worse off.

This is a physical instance of a failure the wider machine-learning literature has begun to document. Shumailov et al. [20] showed that training on model-generated data drives progressive degradation, and Juwara et al. [10] found synthetic augmentation beneficial only under conditions that must be checked rather than assumed. Our setting differs from the recursive loop those studies analyse – we augment once, from a generator trained on real data – which makes the degradation here the more surprising, since a single generation gives error no opportunity to compound. What supplies the damage instead is the absence of a physical constraint: the generator is free to violate the single-valuedness the underlying Maxwell problem guarantees.

5.3 What the surrogate learned

A surrogate that is accurate but opaque is of limited use in design, where the question is not

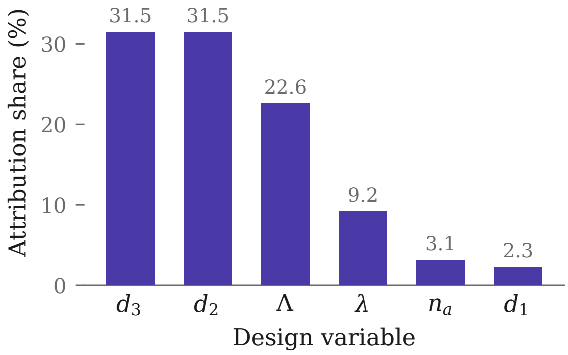

Fig. 2 Global Shapley attribution for the trained SVR, expressed as each variable's share of total attributed magnitude. The outer cladding families d_2 and d_3 jointly account for 63% of the attribution; the innermost holes d_1 are nearly inert.

only what the loss will be but which parameter to change. We applied Shapley additive explanations [15] to the trained SVR, which decomposes each prediction into additive per-feature contributions.

Figure 2 gives the resulting global ranking. The outer cladding diameters d_2 and d_3 dominate, together accounting for 63% of total attribution, followed by the lattice pitch Λ at 23%. Wavelength contributes 9%, and the innermost holes d_1 barely register at 2%.

This ordering is physically consistent, and reading it against the geometry of Fig. 1 explains why. Confinement loss is set by how much of the core field reaches the gold surface, and the outer hole families are what that field must traverse to get there. Enlarging d_2 or d_3 raises the index contrast of the outer cladding, tightening confinement and suppressing leakage. The innermost holes sit inside the field maximum, where they shape the mode profile but do very little to gate its escape – hence their negligible attribution. That the model recovers this ordering from data alone, without being told the physics, is evidence it has learned the leakage mechanism rather than a correlation.

5.4 Manufacturing tolerance

Fiber drawing does not reproduce a design exactly, so a surrogate is only useful for design if it can also say which deviations matter. We perturbed the geometric inputs with Gaussian noise at 1%, 3% and 5% relative standard deviation, ran 100 Monte Carlo trials at each level, and recorded the

Table 5 Monte Carlo scatter of the predicted resonance wavelength under geometric perturbation. Tolerance levels run across the columns, so the growth of each statistic with fabrication error can be read along a row.

Statistic	1%	3%	5%
Peak jitter (std)	1.03	1.27	1.34
Peak spread (range)	2.96	3.00	3.00

100 trials per level, all four geometric variables perturbed simultaneously. Values are in the native wavelength units of the source dataset.

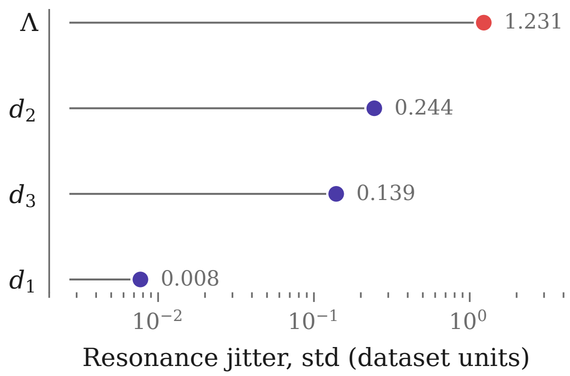

Fig. 3 Resonance jitter attributable to each geometric variable at 3% tolerance, on a logarithmic scale. The lattice pitch (highlighted) dominates by more than a factor of five over the next variable and by two orders of magnitude over the innermost holes.

induced scatter in the predicted resonance wavelength (Table 5). We then repeated the experiment perturbing one variable at a time to attribute the scatter (Fig. 3).

The pitch Λ dominates, contributing over 90% of the observed jitter, while d_1 contributes almost none. The practical reading is a fabrication priority: pitch uniformity along the drawn fiber governs whether a device matches its calibration, whereas tolerance on the smallest cladding holes can be relaxed without meaningful spectral cost.

Two caveats attach to these numbers. First, the peak spread in Table 5 nearly equals the width of the wavelength grid itself, which indicates the peak-detection step is reaching the grid boundary in a substantial fraction of trials; the jitter figures should therefore be read as an upper bound and as a ranking rather than as calibrated magnitudes. Second, the analysis perturbs the surrogate's inputs, so it propagates geometric

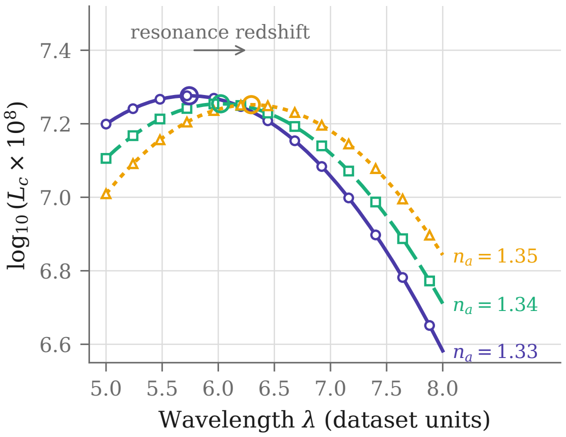

Fig. 4 Predicted confinement-loss spectra at fixed geometry for the three analyte indices. Rings mark the predicted resonance. The peak moves monotonically to longer wavelengths as n_a increases, in uniform steps of 0.276 dataset units per 0.01 refractive index unit.

tolerance but not the surrogate's own predictive error.

5.5 Spectral validation

Aggregate error statistics can hide a surrogate that is right on average and wrong in shape. As a physical check we swept the trained SVR across the full wavelength range at fixed geometry for each analyte and examined the predicted spectra directly (Fig. 4).

The surrogate reproduces the expected behaviour: a single well-formed resonance that shifts monotonically to longer wavelengths as the analyte index rises from 1.33 to 1.35, in near-uniform steps of 0.279 and 0.273 dataset units. Since the model was never given the phase-matching condition, recovering both the direction and the regularity of the shift indicates it has captured the underlying coupling rather than memorising the training spectra. The corresponding sensitivity is 27.63 per refractive index unit in the units of the source dataset.

6 Discussion and limitations

The results support a narrower claim than "kernels beat deep learning". Deep surrogates are not weak here because they are deep; they are weak because nine geometries do not constrain a global function approximator, and the field's response – synthesising more data – addresses the symptom. Where enough real simulation data exists, the neural result of [4] is the better model. The regime we address is the common one where it does not.

Three limitations bound the work. First, everything rests on FV-FEM output; the surrogate inherits whatever the numerical model gets wrong, and experimental validation on a drawn fiber remains necessary. Fabrication effects outside the model – gold surface roughness, material impurity, ellipticity of the drawn holes – will affect sensitivity and signal-to-noise in ways no amount of surrogate accuracy can anticipate.

Second, the surrogate is interpolative within the sampled design space of Table 1. It is not a substitute for FV-FEM outside those ranges, and the LOGO protocol measures generalisation to an unseen geometry within the sampled envelope, not beyond it.

Third, the augmentation paradox is demonstrated for one generator architecture on one dataset. The mechanism we identify – a generator with no physical constraint emitting inconsistent near-duplicate pairs – is not specific to WGAN-GP, and we expect it to affect any unconstrained generative augmentation feeding a kernel model. Establishing that requires testing generators constrained to respect the governing equations, which would be the natural way to obtain the data volume without the inconsistency, and a check on the consistency of generated samples – of the kind performed in Sect. 5.2 – should in our view precede any use of synthetic data in this setting [10].

7 Conclusion

We have shown that a Bayesian-optimised support vector regressor predicts PCF-SPR confinement loss more accurately than a GAN-augmented deep network on identical data – a 61% error reduction – using only the 432 original FV-FEM samples and no synthetic data. Under leave-one-geometry-out validation the advantage holds in both mean and variance.

The augmentation step that the surrogate literature has converged on is, for this class of model, actively harmful. Adding GAN-generated samples raises SVR error by 63% and collapses GPR entirely, and we have traced the collapse to generated sample pairs that are geometrically near-identical yet carry incompatible targets, driving the covariance matrix towards singularity. Generative augmentation exchanges a shortage of data for an inconsistency in it; that exchange is only worth making for a model that cannot function without the volume.

Beyond accuracy, the surrogate yields design guidance. Shapley attribution identifies the outer cladding diameters as the dominant control on confinement loss, consistent with their role in gating evanescent leakage, and Monte Carlo tolerance analysis identifies the lattice pitch as the dominant source of resonance instability under fabrication error. Together these give a designer both the parameter to tune and the parameter to hold, at a computational cost of seconds per evaluation rather than hours.

Supplementary information. The Online Supplement contains the SVR dual formulation and KKT conditions (Sect. S1), the Bayesian optimisation search space and selected hyperparameters (Sect. S2), per-fold leave-one-geometry-out results (Sect. S3), the WGAN-GP configuration and contradiction-search protocol (Sect. S4), a GPR kernel and regulariser ablation (Sect. S5), and the full Monte Carlo tolerance tables (Sect. S6).

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Declarations

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Conflict of interest. The authors declare no competing interests.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The FV-FEM dataset analysed in this study was generated in the course of [23] and is available from the authors on reasonable request.

Materials availability. Not applicable.

Code availability. The model, evaluation and figure-generation code is available from the corresponding author on reasonable request.

Author contribution. M.K.H. designed and implemented the surrogate models, the augmentation experiments and the analyses, and drafted the manuscript. C.K. supervised the machine-learning methodology and revised the manuscript. A.Y. provided the photonic modelling and FV-FEM dataset and revised the manuscript. All authors read and approved the final manuscript.

Prior publication. A preliminary version of this work was presented at the 34th IEEE Signal Processing and Communications Applications Conference (SIU 2026) [8]. The present article substantially extends that paper as described in Sect. 1; all figures and tables have been newly prepared for this submission.

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